

$X_{i0} \sim X_{ip}$ 自变量.
 $b_0 \sim b_p$ 参数.
 e_i 随机变量

$X_{i0}=1$
 b_0 截距.

General Multiple Linear Regression Model

$$Y_i = b_0X_{i0} + b_1X_{i1} + b_2X_{i2} + \dots + b_pX_{ip} + e_i$$

$$= \sum_{j=0}^p b_jX_{ij} + e_i \quad i = 1, \dots, n$$

and, $e_i \sim N(0, \sigma^2)$

There are p independent variables, $p+1$ parameters

Since Y_i is a linear function of the normally distributed independent variable e_i , Y_i is also a normally distributed independent variable, whose distribution is

$$Y_i \sim N\left(\sum_{j=0}^p b_jX_{ij}, \sigma^2\right)$$

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期望.

Multiple linear regression model expressed in matrix form

$$y = Xb + e$$

where

$$y = \begin{bmatrix} Y_1 \\ Y_1 \\ \vdots \\ Y_i \\ \vdots \\ Y_n \end{bmatrix}$$

is a $(n \times 1)$ vector of observed values

$$b = [b_j] = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_j \\ \vdots \\ b_p \end{bmatrix}$$

is a $[(p+1) \times 1]$ vector of regression parameters

参数向量.

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$$X = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{i1} & X_{i2} & \dots & X_{ip} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix}$$

is an $[n \times (p+1)]$ constant matrix of independent variables

$$e = [e_i] = \begin{bmatrix} e_1 \\ \vdots \\ e_i \\ \vdots \\ e_n \end{bmatrix}$$

is an $(n \times 1)$ random vector of normally distributed independent variables,

$$e \sim MVN_n(0, \sigma^2 I)$$

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单位阵.

Since the vector of dependent variables y is a linear function of the vector of normally distributed random variables e , y is also a random vector following normal distribution:

$$y \sim MVN_n(Xb, \sigma^2 I)$$

b, σ^2 are parameters

$$E(y) = E(Xb + e) = E(Xb) + E(e) = Xb$$

$$\text{Var}(y) = \text{Var}(Xb + e) = \text{Var}(Xb) + \text{Var}(e) = \sigma^2 I$$

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Multiple Regression and Variance Analysis

Multiple Regression: study the impact of the independent variables ($X_1, X_2, \dots$) on dependent variable (Y)

$$Y = b_0X_0 + b_1X_1 + b_2X_2 + \dots + b_pX_p + e$$

ANOVA model is a special multiple regression model whose independent variables are 0 or 1

The Analysis of Variance: study the causes (main factors A, B, Interaction effect: $A \times B, \dots$) that cause the variation of dependent variable (Y).

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$$

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The model for variance analysis and Estimation of Parameters

Randomized-Block Experimental Design Model:

Block	Treatment			
Block I	T4	T1	T2	T3
Block II	T1	T4	T3	T2
Block III	T3	T1	T4	T2

Block	Treatment				
Day one	A1	A4	A3	A5	A2
Day two	A4	A1	A5	A2	A3
Day three	A2	A5	A1	A3	A4

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Example The symbols and the data of the boll number per plant in the complete randomized-block experiment of the cotton with four varieties and three replications:

Treatment	Block	Boll Number
1	1	19.44
1	2	24.14
1	3	17.36
2	1	17.24
2	2	19.6
2	3	26
3	1	13.55
3	2	8.35
3	3	13.81
4	1	21.54
4	2	19.94
4	3	23.33

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The linear model of the observed value Y_{ij} of the i^{th} treatment in the j^{th} randomized block can be expressed as the following equation:

$$Y_{ij} = \mu + T_i + B_j + e_{ij}, \quad i = 1, 2, \dots, t$$

where $j = 1, 2, \dots, b$

μ is the population mean
 T_i is the effect of the i^{th} treatment
 B_j is the effect of the j^{th} randomized block
 e_{ij} is the residual effect, and is an independent normally distributed random variable
 μ, T_i and B_j are all unknown parameters

$$e_{ij} \sim N(0, \sigma_e^2)$$

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The linear model of the experimental design

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ Y_{31} \\ Y_{32} \\ Y_{33} \\ Y_{41} \\ Y_{42} \\ Y_{43} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ T_1 \\ T_2 \\ T_3 \\ T_4 \\ B_1 \\ B_2 \\ B_3 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \\ e_{31} \\ e_{32} \\ e_{33} \\ e_{41} \\ e_{42} \\ e_{43} \end{bmatrix}$$

$y = Xb + e$

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If the effects of treatments and randomized blocks are all fixed, then the dependent variable Y_{ij} is normally distributed

$$Y_{ij} \sim N(\mu + T_i + B_j, \sigma_e^2)$$

Expression in Matrix Form:
 Suppose that there are four treatments ($i=1\sim 4$) and three randomized blocks ($j=1,2,3$) in some experiment. Then the model in matrix form is

$$y = Xb + e \sim MVN_{12}(Xb, \sigma_e^2 I)$$

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4.2 Estimators of the Regression Parameters and the Best Linear Unbiased Estimation

$$y = Xb + e \sim MVN(Xb, \sigma_e^2 I)$$

Using least square method: minimizing the residual sum of square

$$Q = e^T e = (y - Xb)^T (y - Xb)$$

$$\frac{\partial Q}{\partial b} = \frac{\partial (y - Xb)^T}{\partial b} (y - Xb) + \frac{\partial (y - Xb)^T}{\partial b} (y - Xb) = -2X^T (y - Xb)$$

Set $\frac{\partial Q}{\partial b} = 0$

$$\Rightarrow 2X^T y = 2X^T Xb$$

$$\Rightarrow b = (X^T X)^{-1} X^T y$$

求偏导时利用了以下标量对向量偏导公式:

$$\frac{\partial}{\partial x} (u^T v) = \left(\frac{\partial}{\partial x} u^T \right) v + \left(\frac{\partial}{\partial x} v^T \right) u$$

$$\frac{\partial x^T}{\partial x} = \frac{\partial x}{\partial x} = I$$

u, v, x 都是向量

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4.2 Estimators of the Regression Parameters and the Best Linear Unbiased Estimation

$$y = Xb + e \sim MVN(Xb, \sigma_e^2 I) \quad b = (X^T X)^{-1} X^T y = Ay$$

If $X^T X$ is a non-singular matrix, then the estimators of the regression coefficients are the only solution of the equations where

$$X^T X = \begin{bmatrix} \sum x_{1i}^2 & \sum x_{1i}x_{2i} & \dots & \sum x_{1i}x_{pi} \\ \sum x_{2i}x_{1i} & \sum x_{2i}^2 & \dots & \sum x_{2i}x_{pi} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_{pi}x_{1i} & \sum x_{pi}x_{2i} & \dots & \sum x_{pi}^2 \end{bmatrix} \quad X^T y = \begin{bmatrix} \sum x_{1i}y_i \\ \sum x_{2i}y_i \\ \vdots \\ \sum x_{pi}y_i \end{bmatrix}$$

$X^T X$ Inverse, Parameter Estimates, and SSE

Variable	Intercept	Height	Age	Weight
Intercept	8.4104038001	-0.156068369	0.1029632577	-141.2237
Height	-0.156068369	0.0081873249	-0.017244935	3.5970265
Age	0.1029632577	-0.017244935	0.0729984165	1.2783325
Weight	-141.2237635	3.5970265113	1.2783325135	2120.099

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$\hat{\mathbf{b}}$ is a normally distributed random vector

$$\hat{\mathbf{b}} \sim MVN_{p+1}(\mathbf{b}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$$

$$E(\hat{\mathbf{b}}) = E[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^TE(\mathbf{y})$$

$$= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\mathbf{b}$$

$$= \mathbf{b}$$

$$Var(\hat{\mathbf{b}}) = Var[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}]$$

$$= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^TVar(\mathbf{y})(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}$$

$$= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\sigma^2\mathbf{I}\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}$$

$$= \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}$$

$$= \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$$

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$$Var(\hat{\mathbf{b}}) = \begin{bmatrix} Var(\hat{b}_0) & Cov(\hat{b}_1, \hat{b}_0) & \cdots & Cov(\hat{b}_p, \hat{b}_0) \\ Cov(\hat{b}_1, \hat{b}_0) & Var(\hat{b}_1) & \cdots & Cov(\hat{b}_1, \hat{b}_p) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(\hat{b}_p, \hat{b}_0) & Cov(\hat{b}_p, \hat{b}_1) & \cdots & Var(\hat{b}_p) \end{bmatrix}$$

$$= \begin{bmatrix} \sigma^2(\hat{b}_0) & \sigma(\hat{b}_1, \hat{b}_0) & \cdots & \sigma(\hat{b}_p, \hat{b}_0) \\ \sigma(\hat{b}_1, \hat{b}_0) & \sigma^2(\hat{b}_1) & \cdots & \sigma(\hat{b}_1, \hat{b}_p) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma(\hat{b}_p, \hat{b}_0) & \sigma(\hat{b}_p, \hat{b}_1) & \cdots & \sigma^2(\hat{b}_p) \end{bmatrix}$$

$\hat{\mathbf{b}}$ is the best linear unbiased estimation (BLUE) of the regression parameters.

最佳无偏估计

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二次型与 $\mathbf{X}^T\mathbf{A}\mathbf{X}$

$\hat{\mathbf{y}}$ is a linear function of the estimated vector of the regression parameters

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{b}}$$

$$= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$= \mathbf{P}_X\mathbf{y}$$

$$\sim MVN_n(\mathbf{X}\mathbf{b}, \sigma^2\mathbf{P}_X)$$

$\hat{\mathbf{y}}$ is a linear function of the vector of observed values, it is an unbiased prediction of $\mathbf{y}$, namely, $E(\hat{\mathbf{y}}) = E(\mathbf{y})$

Residual Vector $\hat{\mathbf{e}}$: the difference between the vector of observed values $\mathbf{y}$ and its estimated vector

$$\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}}$$

$$= \mathbf{y} - \mathbf{P}_X\mathbf{y}$$

$$= (\mathbf{I} - \mathbf{P}_X)\mathbf{y}$$

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The residual vector $\hat{\mathbf{e}}$ is an unbiased estimation of the vector of random errors

$$E(\hat{\mathbf{e}}) = E(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}) = E(\mathbf{y}) - \mathbf{X}E(\hat{\mathbf{b}})$$

$$= \mathbf{X}\mathbf{b} - \mathbf{X}\mathbf{b}$$

$$= \mathbf{0}$$

The Residual Sum of Squares is the product of the transposed residual vector $\hat{\mathbf{e}}^T$ and the residual vector $\hat{\mathbf{e}}$

$$SSE = \hat{\mathbf{e}}^T\hat{\mathbf{e}}$$

$$= (\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}})$$

$$= (\mathbf{y} - \mathbf{X}\hat{\mathbf{b}})^T(\mathbf{y} - \mathbf{X}\hat{\mathbf{b}})$$

$$= \mathbf{y}^T\mathbf{y} - \hat{\mathbf{b}}^T\mathbf{X}^T\mathbf{y}$$

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The residual sum of squares can be further expressed by the quadratic form of dependent variable $\mathbf{y}$

The Residual Mean Square is an unbiased estimation of the random errors

$$\hat{\sigma}^2 = MSE = SSE/(n-p-1)$$

Quadratic Form

$$E(MSE) = \frac{1}{n-p-1} E[\mathbf{y}^T(\mathbf{I} - \mathbf{P}_X)\mathbf{y}]$$

$$= \frac{1}{n-p-1} [\text{tr}[(\mathbf{I} - \mathbf{P}_X)(\sigma^2\mathbf{I})] + \mathbf{b}^T\mathbf{X}^T(\mathbf{I} - \mathbf{P}_X)\mathbf{X}\mathbf{b}]$$

$$= \sigma^2$$

Degree of Freedom

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So $\frac{SSE}{\sigma^2}$ is a χ^2 distribution with degree of freedom of $(n-p-1)$

We can estimate the variance and covariance matrix of the estimated vector of the regression parameters by the residual mean squares

$$Var(\hat{\mathbf{b}}) = \sigma^2(\hat{\mathbf{b}}) = \sigma^2 \cdot (\mathbf{X}^T\mathbf{X})^{-1}$$

$$S^2(\hat{\mathbf{b}}) = MSE(\mathbf{X}^T\mathbf{X})^{-1}$$

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Example (Estimating the Regression Parameters)
Model Weight = Height Age;

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	7215.63710	3607.81855	27.23	<.0001
Error	16	2120.09974	132.50623		
Corrected Total	18	9335.73684			

P-Value

Covariance of Estimates			
Variable	Intercept	Height	Age
Intercept	1114.4309313	-20.68003179	13.643273484
Height	-20.68003179	0.8198591212	-2.285061419
Age	13.643273484	-2.285061419	9.6727452527

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☆ 平方和分解

§4.3 The Analysis of Variance and Decomposition of Sum of Squares

We can further decompose the total sum of squares $SSTO$ into the regression sum of squares SSR and the residual sum of squares SSE

$$SSTO = SSR + SSE$$

Each sum of squares can be expressed by the quadratic form of dependent variable Y

$$\sum_{i=1}^n y_i^2 = y^T y$$

$$\sum_{i=1}^n y_i = y^T \mathbf{1} = \mathbf{1}^T y$$

$$SSTO = \sum_{i=1}^n y_i^2 - \frac{1}{n} \left(\sum_{i=1}^n y_i \right) \left(\sum_{i=1}^n y_i \right)$$

$$= y^T y - \frac{1}{n} y^T \mathbf{1} \mathbf{1}^T y = y^T y - n \bar{y}^2$$

$$= y^T \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) y$$

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$$SSTO = y^T \left[\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right] y$$

$$SSE = y^T y - \hat{b}^T X^T y$$

$$= y^T y - y^T P_X y$$

$$= y^T (\mathbf{I} - P_X) y$$

$$SSR = SSTO - SSE$$

$$= y^T (\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T - \mathbf{I} + P_X) y$$

$$= y^T (P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T) y$$

$\hat{e} = y - \hat{y}$

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Two different definitions of $SSTO$

$y = Xb + e \Rightarrow \hat{b} = (X^T X)^{-1} X^T y$
 $\hat{y} = X\hat{b} = X(X^T X)^{-1} X^T y = P_X y$

No correction by mean
 $SSTO = y^T y$

Correction by mean
 $SSTO = y^T y - n \bar{y}^2$
 $= y^T \left[\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right] y$

$SSR = \hat{y}^T \hat{y} = (X\hat{b})^T X\hat{b}$
 $= \hat{b}^T X^T X (X^T X)^{-1} X^T y$
 $= \hat{b}^T X^T P_X y$
 $= \hat{b}^T X^T y = y^T P_X y$

$SSE = \hat{e}^T \hat{e}$
 $= (y - \hat{y})^T (y - \hat{y})$
 $= y^T y - \hat{b}^T X^T y$

$SSR = \hat{y}^T \hat{y} - n \bar{y}^2$
 $= \hat{b}^T X^T y - n \bar{y}^2$

$SSE = (y - \hat{y})^T (y - \hat{y})$
 $= y^T y - \hat{b}^T X^T y$

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☆ The expectation of the regression sum of squares SSR is inferred by the following,

$$E(SSR) = E(y^T (P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T) y)$$

$$= \sigma^2 \text{tr} \left(P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) + b^T X^T (P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T) X b$$

$$= p\sigma^2 + b^T (X^T X - \frac{1}{n} X^T \mathbf{1} \mathbf{1}^T X) b$$

利用了: $\text{tr}(P_X) = \text{rank}(X) = p + 1$, $\text{tr}(\frac{1}{n} \mathbf{1} \mathbf{1}^T) = \frac{1}{n} \text{tr}(\mathbf{1} \mathbf{1}^T) = 1$

Decompose the degrees of freedom:

Total Degrees of Freedom $(n-1)$ =
Regression Degrees of Freedom p + Residual Degrees of Freedom $(n-p-1)$

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The Expectation of mean-squares in the regression

✓ $MSR = SSR/p$
 $E(MSR) = E(SSR)/p$

$$= \sigma^2 + \frac{1}{p} b^T (X^T X - \frac{1}{n} X^T \mathbf{1} \mathbf{1}^T X) b$$

$$E(MSR) = \sigma^2 + \frac{1}{p} \left(\sum_{j=1}^p b_j^2 SS_j + 2 \sum_{j=1}^{p-1} \sum_{k=j+1}^p b_j b_k SCP_{jk} \right)$$

where $SS_j = \sum_{i=1}^n (X_{ij} - \bar{X}_j)^2$, $SCP_{jk} = \sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_k)$

If the regression parameters $b_i (i=1, 2, \dots, p)$ are all zero, then

$$E(MSR) = \sigma^2$$

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方差分析表

Table of the analysis of variance of multivariate linear regression model

Source	Degrees of Freedom	Sum of Squares	Mean-Squares	Expected Mean-Squares
Regression	p	SSR	MSR	$\sigma^2 + \frac{1}{p}b^T(X^T X - \frac{1}{n}X^T 11^T X)b$
Residual Errors	$n-p-1$	SSE	MSE	σ^2
Total Sum	$n-1$	$SSTO$		

当所有 $b_i = 0$ ($i=1, \dots, p$) 时,

31 $E(MSR) = E(MSE)$. 说明统计量

$F = \frac{MSR}{MSE}$ 在 1 附近 (大概率区域). $\rightarrow H_0$ 接受 $\rightarrow$ 无明显线性联系.

变量加多, 导致 $R^2 \downarrow$.

Coefficient of Multiple Determination (多元决定系数):

The ratio of the multiple regression sum of squares SSR to the total sum of squares $SSTO$

$$R^2 = SSR/SSTO = 1 - SSE/SSTO$$

It measures the proportion of total variation which can be explained by p independent variables in model

Adjusted Coefficient of Multiple Determination (调整多元决定系数)

$$R_a^2 = 1 - MSE/MSTO$$

$$MSE = SSE/(n-p-1)$$

$$MSTO = SSTO/(n-1)$$

Adding some irrelevant independent variables into the regression model may cause decrease of the adjusted coefficient of multiple determination.

So the ratio of the regression sum of squares to the random-error sum of squares follows an F distribution

$$F^* = \frac{SSR/(p\sigma^2)}{SSE/[(n-p-1)\sigma^2]} = MSR/MSE \sim F_{(p, n-p-1)} \quad H_0$$

We can test whether there's a linear regression relationship between the dependent variable Y and the independent variables X_i ($i=1, 2, \dots, p$) by F test.

Large F test statistic over the critical value only indicates that there may be a linear regression relationship between Y and the independent variable X_i ($i=1, 2, \dots, p$), but it doesn't mean that all regression coefficients reached significant level, it maybe due to the significance of several of coefficients.

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Example (Analysis of Variance of Multiple Regression)

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	7215.63710	3607.81855	27.23	<.0001
Error	16	2120.09974	132.50623		
Corrected Total	18	9335.73684			

Root MSE = 11.51114
Dependent Mean = 100.02632
Coeff Var = 11.50811
R-Square = 0.7729
Adj R-Sq = 0.7445
CV = SD + Mean * 100%
= 11.51114 + 100.02632 * 100%

Variation Coefficient

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Example: We investigated six agronomic traits of 16 genetic materials in some cotton experiment, the average of three blocks are calculated for the flowering number per plant until Aug. 9th (A), the flowering number per plant until Sep. 3rd (S), the boll number per plant (B), the weight per plant (W), lint (L) and yield (Y), respectively. The observed values of the six traits are as follows

	1/3 Flowering Number	2/3 Flowering Number	Boll Number	Weight (g)	Lint (%)	Yield (kg)
1	17.02	28.4	11.5	4.98	41.39	64.38
2	14.08	25.38	11.05	5.33	40.21	62.5
3	20.3	26.56	10.57	5.2	38.68	57.02
4	19.71	32.18	14.33	5.4	40.27	57.18
5	16.32	31.33	14.12	6.1	40.28	79.66
6	16.34	30.71	14.21	4.89	36.84	41.2
7	19.69	27.14	11.76	5.54	39.22	53.3
8	18.45	32.78	16.84	6.01	40.91	66
9	20.93	36.86	10.76	4.84	38.11	50.55
10	18.94	26.93	13.25	5.11	39.09	53.62
11	15.2	31.77	7.01	4.57	36.7	35.16
12	19.18	26.34	12.25	5.16	37.77	64.72
13	18.2	30.07	13.2	5.21	39.46	64.72
14	18.33	31.7	16.63	5.9	39.57	74.7
15	19.03	26.98	14.14	5.28	37.82	48.04
16	16.74	29.66	13.5	4.36	38.16	39.72



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data cotton;
input f1 f2 b w l y @@;

```
cards;
17.82 29.4 11.5 4.98 41.39 58.38
14.08 25.38 12.05 5.33 40.21 62.5
20.3 26.56 10.57 5.2 38.68 57.02
19.71 32.18 14.33 5.4 40.27 57.18
16.32 31.33 14.12 6.1 40.28 79.66
16.34 30.71 14.21 4.89 36.84 41.2
19.69 27.14 11.76 5.54 39.22 53.3
18.45 32.78 16.84 6.01 40.91 66
20.93 26.86 10.76 4.84 38.11 50.55
18.94 26.93 13.25 5.11 39.09 53.62
15.2 21.77 7.01 4.57 36.7 35.16
19.18 26.34 12.25 5.16 37.77 64.72
18.8 30.07 13.2 5.21 39.46 64.72
18.33 31.7 16.63 5.9 39.57 74.7
19.03 26.98 14.14 5.28 37.82 48.04
16.74 29.66 13.5 4.36 38.16 39.72
;
```

```
proc reg;
model y = b w l;
run;
```

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If the yield of cotton (Y) is determined by three factors which are boll number per plant (B), weight per boll (W) and lint (L), then the multiple linear regression model between the cotton's yield and the three independent variables is

$$Y_i = b_0 + b_B B_i + b_W W_i + b_L L_i + e_i$$

The analysis result of multiple regression variance of this model is listed in the following table.

Table of the analysis of variance of the ternary linear regression of the cotton's yield

Source	Degrees of Freedom	Sum of Squares	Mean-Squares	F value	P value
Regression	3	1785.29	595.1	18.97	0.0001
Residual Error	12	376.51	31.38		
Total Sum	15	2161.8			

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n-1

Value of the coefficient of multiple determination

$$R^2 = SSR/SSTO = 1785.29/2161.80 = 0.8258$$

Value of the adjusted coefficient of multiple determination

$$R_a^2 = 1 - MSE / MSTO = 1 - 31.38 / 141.79 = 0.7823$$

The result indicates that the F test has reached the level of significance, so the null hypothesis is rejected

$$H_0: b_B = b_W = b_L = 0$$

So there's a regression relationship between the cotton's yield (Y) and three independent variables which are boll number per plant (B), weight per boll (W) and lint (L). More than 80% of the variation of the cotton's yield is determined by the three factors.

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b_i 的显著性检验

The statistical test for the parameter of i -th independent variable is based on t distribution, the statistic is

$$t^* = \hat{b}_i / SE(\hat{b}_i) \sim t(n-p-1)$$

Since $\hat{b} \sim MVN_{p+1}(\mathbf{b}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$, $SE(\hat{b}_i)$ is the root square of the i -th diagonal element of the matrix $MSE(\mathbf{X}^T\mathbf{X})^{-1}$

Table 4.3 The analysis result of the ternary regression parameters of cotton's yield

Variable	Estimator	F Value	P Value
Intercept	-147.46	-3.48	0.005
Boll Number Per Plant	-0.12	-0.16	0.878
Weight Per Boll (W)	16.57	3.8	0.003
Lint (L)	3.03	2.31	0.039

The analysis result indicates that the regression parameter of boll number per plant (B) doesn't reach the significant level of 5%.

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Thus, it is not suitable to predict the cotton's yield by the following regression equation

$$\hat{Y} = -147.46 - 0.12B + 16.57W + 3.03L$$

So this multiple linear regression model still needs to be improved.

Generalized Linear Test $H_0: \mathbf{Kb} - \mathbf{m} = 0$

Proc REG;

Model Yield = Boll Weight Lint;

Mtest Boll, Weight, Lint; $H_0: b_B = b_W = b_L = 0$

Mtest Weight; $H_0: b_W = 0$

Mtest Boll, Lint; $H_0: b_B = b_L = 0$

Mtest Boll, Weight - 17, Lint - 3; $H_0: b_B = 0, b_W = 17, b_L = 3$

Mtest Weight - 5 * Lint; $H_0: b_W = 5b_L$

Run;

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Generalized Linear Test $H_0: \mathbf{Kb} - \mathbf{m} = 0$

How to determine the $\mathbf{K}, \mathbf{m}$

Example

$$H_0: b_B = 0, b_W = 17, b_L = 3$$

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} b_0 \\ b_B \\ b_W \\ b_L \end{bmatrix} = \begin{bmatrix} 0 \\ 17 \\ 3 \end{bmatrix}$$

(3×4) matrix

Determine all elements of $\mathbf{K}$ according to the equations of H_0

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§ 4.4 Interval Estimation of Regression Parameters and Prediction

We can get the confidential interval of a regression parameter b_j by the following t distribution,

$$(\hat{b}_j - b_j) / SE(\hat{b}_j) \sim t_{(n-p-1)}$$

where $SE(\hat{b}_j)$ is an unbiased estimate of $\sigma(\hat{b}_j)$

$$SE(\hat{b}_j) = \sqrt{MSE[\mathbf{X}^T\mathbf{X}]^{-1}_{jj}}$$

The $100(1-\alpha)\%$ confidential interval of the regression parameter b_j is determined by the following formula

$$\hat{b}_j - t_{(\alpha/2, n-p-1)} SE(\hat{b}_j) \leq b_j \leq \hat{b}_j + t_{(\alpha/2, n-p-1)} SE(\hat{b}_j)$$

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$$\hat{y} = \hat{b}_0 + \hat{b}_1 X_1 + \dots + \hat{b}_p X_p$$

给定一个新的自变量向量 X_0 , 如何来预测 Y , 平均预测, 单次预测的置信区间

This confidence interval can be used to infer the increase amount of the dependent variable Y when the independent variable X increase one unit with other independent variables kept in the model.

Parameter Estimates				
Variable	DF	Variance Inflation	95% Confidence Limits	
Intercept	1	0	-239.77000	-55.15373
Boll	1	1.61632	-1.79399	1.55269
Weight	1	2.12275	7.06009	26.08961
Lint	1	1.59424	0.17145	5.88313

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The Bonferroni Joint Confidence Interval for g ($\leq p+1$) regression parameters:

$$\hat{b}_j - t_{(\alpha/(2g); n-p-1)} SE(\hat{b}_j) \leq b_j \leq \hat{b}_j + t_{(\alpha/(2g); n-p-1)} SE(\hat{b}_j) \quad j = 1, \dots, g$$

When all the regression parameters are inferred, we can also use F distribution to obtain the boundary of the joint confidential region

$$\frac{(\hat{\mathbf{b}} - \mathbf{b})^T (\mathbf{X}^T \mathbf{X})^{-1} (\hat{\mathbf{b}} - \mathbf{b})}{(p+1)MSE} \leq F_{(1-\alpha; p+1, n-p-1)}$$

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For a group of given independent variables $\mathbf{x}_0^T = (1 \ x_{01} \ \dots \ x_{0p})$

The average response of dependent variables Y is $E(\hat{\mu}_0) = \mu_0 = \mathbf{x}_0^T \mathbf{b}$

The unbiased estimator of the average response is $\hat{\mu}_0 = \mathbf{x}_0^T \hat{\mathbf{b}}$ $E(\hat{\mu}_0) = \mathbf{x}_0^T E(\hat{\mathbf{b}}) = \mathbf{x}_0^T \mathbf{b}$

whose variance is $\sigma^2(\hat{\mu}_0) = \mathbf{x}_0^T \text{Var}(\hat{\mathbf{b}}) \mathbf{x}_0 = \sigma^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$ average

The estimated variance $S^2(\hat{\mu}_0) = MSE \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$

The $100(1-\alpha)\%$ confidence interval of the average response μ_0 is determined by

$$\hat{\mu}_0 - t_{(\alpha/2; n-p-1)} SE(\hat{\mu}_0) \leq \mu_0 \leq \hat{\mu}_0 + t_{(\alpha/2; n-p-1)} SE(\hat{\mu}_0)$$

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$\delta(\hat{\mu}_0)$

When inferring some g average response at the independent variables vector $\mathbf{x}_0$, the Bonferroni Joint Confidence Interval is

$$\hat{\mu}_0 \pm t_{(\frac{\alpha}{2g}; n-p-1)} SE(\hat{\mu}_0)$$

Given the vector of a group of independent variables ($\mathbf{x}_0^T$), we can predict a new observed value Y_0 by $\hat{Y}_0 = \mathbf{x}_0^T \hat{\mathbf{b}}$

The $100(1-\alpha)\%$ predicted interval of the new observed value Y_0 is determined by

$$\hat{Y}_0 - t_{(\alpha/2; n-p-1)} SE(\hat{Y}_0) \leq Y_0 \leq \hat{Y}_0 + t_{(\alpha/2; n-p-1)} SE(\hat{Y}_0)$$

$$SE(\hat{Y}_0) = \sqrt{MSE [1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0]}$$

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Given a group of independent variables $\mathbf{x}_0^T$, we can predict the mean of m new observed values Y_0 , whose predicted upper and lower boundaries are computed by the following formula.

$$\hat{Y}_0 \pm t_{(\alpha/2; n-p-1)} \sqrt{MSE \left[\frac{1}{m} + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0 \right]}$$

When simultaneously predicting g new observed values given different independent variables, the $100(1-\alpha)\%$ Bonferroni joint predicted upper and lower boundaries are

$$\hat{Y}_0 \pm t_{(\frac{\alpha}{2g}; n-p-1)} SE(\hat{Y}_0)$$

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In the example of multiple regression analysis of cotton's yield, if we delete the boll number per plant (B), then the regression parameters of the weight per boll (W) and lint (L) will reach the level of significance of 5% below. Thus, there is a significant regression relationship between the yield of cotton (Y) and the weight per boll (W) and lint (L). The coefficient of multiple determination has reached 82.55%.

The result of parameter estimation for the regression of the cotton yield on weight per boll (W) and Lint (L)

Variable	Estimate	Standard Error	t* value	P value
Interception	-146.68	40.47	-3.63	0.003
Weight Per Boll (W)	16.23	3.63	4.47	0.001
Lint (L)	3.01	1.26	2.4	0.032

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The analysis result indicates that it is suitable to predict the yield of cotton by weight per boll (W) and lint (L),

$$\hat{Y} = -146.68 + 16.23W + 3.01L$$

The 95% confidence intervals of the regression coefficient of weight per boll (W) and lint (L) are

$$16.23 - 2.16 \times 3.63 = 8.39 \leq b_W \leq 24.07 = 16.23 + 2.18 \times 3.63$$

$$3.01 - 2.16 \times 1.26 = 0.26 \leq b_L \leq 5.76 = 3.01 + 2.16 \times 1.26$$

The Bonferroni joint confidential intervals of the two regression parameter are

$$16.23 - 2.53 \times 3.63 = 7.05 \leq b_W \leq 25.41 = 16.23 + 2.53 \times 3.63$$

$$3.01 - 2.53 \times 1.26 = -0.18 \leq b_L \leq 6.20 = 3.01 + 2.53 \times 1.26$$

Note: $t(0.05; 13) = 2.16$, $t(0.025; 13) = 2.53$

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The observed values of the first genetic material are the boll number per plant $B=11.5$ /plant, lint $L=41.39\%$, and the cotton yield $Y=58.38$ kg.

Confidence interval for the average estimate of cotton yield

$$50.52 \leq \mu_0 \leq 67.24$$

Confidence interval for the new observed value of cotton yield

$$44.55 \leq Y_0 \leq 73.21$$

The observed values of the tenth genetic material are the boll number per plant $B=13.35$ /plant, lint $L=39.09\%$, and the cotton yield $Y=53.62$ kg.

The confidence interval for the average estimator of cotton yield

$$50.94 \leq \mu_0 \leq 57.19$$

The confidence interval for the new observed value of cotton yield

$$42.01 \leq Y_0 \leq 66.11$$

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```
*Proc REG;
  Model Yield = Weight Lint/ALL;
  Mtest Weight, Lint;
  Mtest Weight;
  Mtest Lint;
  Mtest Weight -17, Lint -3;
  Mtest Weight-5 * Lint;
run;
```

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-146.68158	40.46502	-3.62	0.0031
Weight	1	16.23032	3.63185	4.47	0.0006
Lint	1	3.01373	1.25787	2.40	0.0323

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Parameter Estimates

Variable	DF	95% Confidence Limits
Intercept	1	-234.10094 -59.26223
Weight	1	8.38418 24.07645
Lint	1	0.29627 5.73120

Consistent Covariance of Estimates

Variable	Intercept	Weight	Lint
Intercept	719.53541341	-15.69933929	-16.47653904
Weight	-15.69933929	8.9368348164	-0.740178152
Lint	-16.47653904	-0.740178152	0.5185898261

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Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	95% CL Mean	95% CL Predict
1	58.3800	58.8898	3.8702	50.5228 67.2448	44.5536 73.2141
2	62.5000	61.0082	1.8826	58.9412 65.0753	49.6798 73.3387
3	57.0200	54.2873	1.3950	51.2714 57.3032	42.2646 66.3100
4	57.1800	62.3252	1.8677	58.2902 66.3592	50.0072 74.6431
5	79.6600	73.7165	2.8333	67.5955 79.8376	60.5667 86.8683
6	41.2000	43.7106	2.5983	38.0974 49.3238	30.7894 56.6318
7	53.3000	61.4330	1.6505	57.8673 64.9988	49.2607 73.6053
8	66.0000	74.1545	2.6752	68.3751 79.3333	61.1602 87.1487
9	50.5500	46.7865	1.7978	42.8427 50.6104	34.4573 58.9958

Output Statistics

Obs	Std Error Residual	Student Residual	-2	-1	0	1	2	Cook's D
1	3.747	-0.134						0.006
2	5.048	0.296						0.004
3	5.203	0.525						0.007
4	5.053	-1.018						0.047
5	4.582	1.297						0.214
6	4.719	-0.532						0.029
7	5.128	-1.586						0.087
8	4.676	-1.744						0.332
9	5.078	0.753						0.024

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§ 4.5 Multiple Regression Model Through the Origin

The multiple regression model through the origin is suitable for:

1. The dependent variable Y is zero when all the independent variables are zero;
2. Statistical test indicates that the regression interception is zero.

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$$Y_i = b_1 X_{i1} + b_2 X_{i2} + \dots + b_p X_{ip} + e_i$$

$$= \sum_{j=1}^p b_j X_{ij} + e_i \quad i = 1, \dots, n$$

where

$b_1, b_2, \dots, b_p$ the slope of the regression model, unknown parameters

$X_{i1}, X_{i2}, \dots, X_{ip}$ independent variables of the regression model, which are given constants

e_i is an independent random variable, who has an independent normal distribution $e_i \sim N(0, \sigma^2)$

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Matrix form of the regression model

$$y = Xb + e$$

where

y is an $(n \times 1)$ vector of observed values

b is a $(p \times 1)$ vector of the regression parameters

X is an $(n \times p)$ constant matrix of independent variables

e is an $(n \times 1)$ random vector following multivariate normal distribution

$$e \sim MVN_n(0, \sigma^2 I)$$

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$$\hat{b} = (X^T X)^{-1} X^T y$$

where

$$X^T X = \begin{bmatrix} \sum X_{i1}^2 & \sum X_{i1} X_{i2} & \dots & \sum X_{i1} X_{ip} \\ \sum X_{i1} X_{i2} & \sum X_{i2}^2 & \dots & \sum X_{i2} X_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ \sum X_{ip} X_{i1} & \sum X_{ip} X_{i2} & \dots & \sum X_{ip}^2 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} \sum X_{i1} Y_i \\ \sum X_{i2} Y_i \\ \vdots \\ \sum X_{ip} Y_i \end{bmatrix}$$

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Table of the analysis of variance of multiple linear regression through the origin

Sources	Degrees of Freedom	Sum of Squares	Mean-Squares	Expected Mean-Squares
Regression	p	$SSR = y' P_x y$	MSR	$\sigma^2 + \frac{1}{p} b' (X' X) b$
Residual	$n-p$	$SSE = y' (I - P_x) y$	MSE	σ^2
Total	n	$SSTO = y' y$		

We still take the multiple linear regression model in example 4.1 about the relationship of the cotton yield (Y) and its three component factors which are the boll number per plant (B), the weight per boll (W) and lint (L) as an example. Let's compare the analysis efficiency between the models through and not through the origin

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If the factors that make up of the yield are all zero, then the cotton yield must be zero. So we should adopt the triple variables regression model through the origin in the analysis,

$$Y_i = b_B B_i + b_W W_i + b_L L_i + e_i$$

Proc REG;

Model Yield = Boll Weight Lint/NOINT;

Table 4.6 the variance analysis of the linear regression model through the origin with triple variables for the cotton yield

Sources	Degrees of Freedom	Sum of Squares	Mean-Squares	F value	P value
Regression	3	51638.31	17212.77	295.74	0.0001
Residual	13	756.62	58.2		
Total	16	52394.93			

The analysis result of variance indicates that the F test has reached the most significant level

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$$Y_i = b_B B_i + b_W W_i + b_L L_i + e_i$$

The coefficient of multiple determination is

$$R^2 = SSR / SSTO = 0.9856$$

The adjusted coefficient of multiple determination is

$$R_a^2 = 1 - MSE / MSTO = 0.9822$$

Nearly above 98% of the variation of the cotton yield is determined by the three factors.

The estimators of the adjusted and non-adjusted coefficient of multiple determination are both larger than that of the three variables regression with inclusion of intercept ($R^2=0.8258$, $R_a^2=0.7823$).

$$Y_i = b_0 + b_B B_i + b_W W_i + b_L L_i + e_i$$

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Analysis of variance of the simple linear regression passing the origin for the cotton yield

(1) Simple regression of the yield of cotton ($\bar{Y}$) on the flowering number per plant until Sep. 3rd (X_1) $\hat{Y} = 197X_1$

Sources	DF	Sum of Squares	Mean-Squares	Coeff. of Deter.
Regression	1	$SSR(X_1) = 50851.25$	$MSR(X_1) = 50851.25$	0.971
Residual	15	$SSE(X_1) = 1543.68$	$MSE(X_1) = 102.91$	

(2) Simple regression of the yield of cotton ($\bar{Y}$) on the boll number per plant (X_2) $\hat{Y} = 4.29X_2$

Sources	DF	Sum of Squares	Mean-Squares	Coeff. of Deter.
Regression	1	$SSR(X_2) = 50609.39$	$MSR(X_2) = 50609.39$	0.966
Residual	15	$SSE(X_2) = 1785.84$	$MSE(X_2) = 119.056$	

Analysis of variance in the bivariate linear regression model passing the origin for the cotton yield

(3) Bivariate regression of the yield of cotton ($\bar{Y}$) on the flowering number per plant until Sep. 3rd (X_1) and the boll number per plant (X_2) $\hat{Y} = 1.40X_1 + 1.24X_2$

Sources	DF	Sum of Squares	Mean-Squares	Coeff. of Deter.
Regression	2	$SSR(X_1, X_2) = 50898.48$	$MSR(X_1, X_2) = 25449.24$	0.971
Residual	14	$SSE(X_1, X_2) = 1496.45$	$MSE(X_1, X_2) = 106.89$	

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Extra sum of squares after the second variables the boll number per plant (X_2), is added into model, with the first independent variable the flowering number per plant (X_1) remained in the model

$$SSR(X_2 | X_1) = SSR(X_1, X_2) - SSR(X_1)$$

$$= 50898.48 - 50851.25$$

$$= 47.23 \quad \text{The extra effect is very small, thus, it is negligible}$$

The variance analysis of linear regression model can not only decompose the total sum squares $SSTO$ into the regression sum of squares SSR and the residual sum of squares SSE , but also doing decompose of SSR and SSE .

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For multiple linear regression model

$$Y_i = b_0 + b_1X_{i1} + b_2X_{i2} + \dots + b_pX_{ip} + e_i$$

Several methods to decompose the regression sum of square with p degree of freedom, for example:

$$SSR(X_1, \dots, X_p) = SSR(X_1) + SSR(X_2 | X_1) + SSR(X_3 | X_1, X_2) + \dots + SSR(X_{p-1} | X_1, \dots, X_{p-2}) + SSR(X_p | X_1, \dots, X_{p-1})$$

In result, p sum of squares with 1 degree freedom each one will be obtained

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The regression sum of squares $SSR(X_1, X_2, \dots, X_p)$ can be decomposed according to the real needs.

If there are two groups of the independent variables ($X_1, X_2, \dots, X_p$), ($X_{q+1}, X_{q+2}, \dots, X_p$), then we can decompose the total regression sum of squares into the regression sum of squares of any two items as follows

$$SSR(X_1, \dots, X_p) = SSR(X_1, \dots, X_q) + SSR(X_{q+1}, \dots, X_p | X_1, \dots, X_q) \\ = SSR(X_{q+1}, \dots, X_p) + SSR(X_1, \dots, X_q | X_{q+1}, \dots, X_p)$$

so that we can test the size of extra regression effect due to one group of independent variables on the dependent Y when the other group of independent variables exists in the model.

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The residual sum of squares $SSE(X_1, X_2, \dots, X_p)$ with degrees of freedom of $(n-p-1)$ can also be partitioned in further. Besides the random error effect, the residual sum of squares SSE may also contain some other effects from unknown variables that are not in the regression model. So we can keep on adding new variables by further decomposition of the residual sum of squares SSE .

$$SSTO = SSR(X_1, \dots, X_p) + SSE(X_1, \dots, X_p) \\ = \overbrace{SSR(X_1, \dots, X_p) + SSR(X_{p+1} | X_1, \dots, X_p)} \\ + SSE(X_1, \dots, X_{p+1}) \\ = SSR(X_1, \dots, X_{p+1}) + SSE(X_1, \dots, X_{p+1})$$

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As the number of independent variables increase, the residual sum of squares SSE will decrease, on contrary, the regression sum of squares will increase.

$$SSTO = SSR(X_1) + SSE(X_1) \\ = SSR(X_1) + SSR(X_2 | X_1) + SSE(X_1, X_2) \\ = SSR(X_1) + SSR(X_2 | X_1) + SSR(X_1 | X_1, X_2) \\ + SSE(X_1, X_2, X_1) \\ = \dots$$

These extra sum of squares each with $df = 1$ is independent of the residual sum of squares.

So F test can be applied to test the extra effects of these new included independent variables on the dependent variable.

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平方和分解, 自由度分解.

$SSTO$ 自由度: $n-1$.

每一项额外平方和自由度: 1. 总共为 p .

$SSE(X_1, X_2, \dots, X_p)$ 自由度为: $n-p-1$

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$$F^* = \frac{SSR(X_i | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_p)}{MSE(X_i, \dots, X_p)} \sim F(1, n-p-1)$$

等价于检验 $H_0: b_i = 0$
t/分布.

Testing whether the extra effect of the independent variable X_2 reach the level of significance, given the independent variable X_1 in the model

$$F^* = SSR(X_2 | X_1) / MSE(X_1, X_2) \sim F(1, n-3)$$

Testing whether the extra effect of the independent variable X_3 reach the level of significance, given the independent variable X_1, X_2 in the model

$$F^* = SSR(X_3 | X_1, X_2) / MSE(X_1, X_2, X_3) \sim F(1, n-4)$$

Type I for F-test

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The extra regression effect can also be measured by the partial coefficient of determination (部分决定系数)

The proportion of the decreased variation in total variation of Y caused by adding X_2 into the regression model when X_1 exists:

$$r_{2.1}^2 = SSR(X_2 | X_1) / SSE(X_1)$$

The proportion of the decreased variation in total variation of Y caused by adding X_3 into the regression model when X_1, X_2 exists:

$$r_{3.12}^2 = SSR(X_3 | X_1, X_2) / SSE(X_1, X_2)$$

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Sometimes, it is rather difficult to select a good combination of independent variables in multiple regression analysis. There are several different statistical methods for selecting independent variables. We may get completely different regression models for the same set of experimental data by different statistical method.

It is very difficult to decide which combination of independent variables is "the best", because maybe there's no best regression model.

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Three methods of choosing the independent variables:

- (1) backward elimination (向后消元法),
- (2) forward selection (向前选择法),
- (3) stepwise regression (逐步回归法).

In SAS, two procedures (Stepwise, Reg) can be used for variable selection

```

Data Cotton;
Input B W L Yield;
BW = B*W; BL = B*L; WL = W*L; BWL = B*W*L;
Datalines;
Proc Stepwise;
Model Yield = B W L BW BL WL BWL;
Noint Backward SLEntry = 0.05;
run;

Proc Stepwise;
Model Yield = B W L BW BL WL BWL;
Noint Forward SLEntry = 0.05;
run;

Proc Stepwise;
Model Yield = B W L BW BL WL BWL;
Noint Stepwise SLEntry = 0.05 SLEstay = 0.05;
run;

```

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Backward Elimination: firstly, all independent variables are included in the regression model, then the unimportant variables, one by one, is excluded according to their extra effects, at last, a reduced model will be obtained, in which all retained independent variables must reach the level of significance.

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The requirement of backward elimination is that the F_k^* value of the deleted variable X_k must be the smallest and the most be insignificant,

$$F_k^* = \frac{SSR(X_k | X_1, \dots, X_p)}{MSE(X_1, \dots, X_p)} = \frac{\hat{\delta}_k^2}{s^2(\hat{\delta}_k)} = (\hat{\delta}_k / s(\hat{\delta}_k))^2 = t^2$$

If there is following multiple regression model of the cotton yield (Y) on its component factors which are the boll number per plant (B), the weight per boll (W) and lint (L)

$$Y_i = b_B B_i + b_W W_i + b_L L_i + b_{BW} BW_i + b_{BL} BL_i + b_{WL} WL_i + b_{BWL} BWL_i + e_i$$

where $BW = B \times W$, $BL = B \times L$, $WL = W \times L$, $BWL = B \times W \times L$.

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Stepwise regression is an improved selection of independent variables, combining the forward selection and the backward elimination, and is the most widely used in selecting independent variables of multivariate regression model.

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This method

1. Add new variables by forward selection;
2. After a new variable is added, the significance of the extra regression effects of other variables in the model are tested. Then those detected insignificant variables are deleted;
3. Continue the selection and the deletion;
4. Until no new variable can be added, and no included variables can be deleted.

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Perform stepwise regression analysis on the cotton data, the same result as that of the forward selection is obtained.

We find two-variable regression models similar in predicting dependent Y by three methods of selecting independent variables on the cotton data. This indicates that we may get several good regression models by rational selection of variables in multiple regression analysis, thus it is difficult to determine unique best regression model.

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Chapter 4: Methods of Multi-variable Regression Analysis

- 4.1 Multiple Regression Model
- 4.2 Estimators of the Regression Parameters and the Best Linear Unbiased Estimation
- 4.3 The Analysis of Variance and Decomposition of Sum of Squares
- 4.4 Interval Estimation of Regression Parameters and Prediction
- 4.5 Multiple Regression Model through the Origin
- 4.6 Selection of Independent Variables

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Summary

$$y = Xb + e$$

- ◆ Assumption: $e \sim MVN(0, \sigma^2 I) \rightarrow y \sim MVN(Xb, \sigma^2 I)$
- ◆ BLUE: $\hat{b} = (X^T X)^{-1} X^T y$, $\hat{b} \sim MVN(b, \sigma^2 (X^T X)^{-1})$
- ◆ BLUP: $\hat{y} = X\hat{b} = P_X y$, $\hat{y} \sim MVN(Xb, \sigma^2 P_X)$
- ◆ $\hat{\sigma}^2 = MSE = (y - \hat{y})^T (y - \hat{y}) = y^T (I - P_X) y$, $df = ?$ $n - p - 1$
- ◆ $SSR = \hat{y}^T \hat{y} - ny^2 = y^T P_X y - ny^2 = ?$ $df = ?$ p
- ◆ $SSTO = y^T y - ny^2 = ?$ $df = ?$ $n - 1$
- ◆ For the model through the origin, need to remove ny^2

Statistical testing

F-test for whole model

F-test (generalized linear test) for single or a group of independent variables

t-test for single independent variable

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